

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<tool_call> reasoning process here

To find the de Broglie wavelength of an electron given its kinetic energy , we use the de Broglie wavelength formula :

$\lambda = h / p$

where :

- h is Planck 's constant = 6.626×10^{-34} J · s
- p is the momentum of the electron

For a non-relativistic electron , kinetic energy (K.E.) is given by :

$K.E. = (1/2)mv^2$

and momentum $p = mv$

So , $p = \sqrt{2mK.E.}$

Thus , de Broglie wavelength becomes :

$\lambda = h / \sqrt{2mK.E.}$

We need to compute this for $K.E. = 1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$

Mass of electron $m = 9.109 \times 10^{-31} \text{ kg}$

Plugging in :

$\lambda = (6.626 \times 10^{-34}) / \sqrt{2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19}}$

First compute the denominator :

$2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19} = 2.918 \times 10^{-49}$

$\sqrt{2.918 \times 10^{-49}} \approx 5.402 \times 10^{-25}$

Then :

$\lambda = 6.626 \times 10^{-34} / 5.402 \times 10^{-25} \approx 1.227 \times 10^{-9} \text{ m}$

Convert to cm : $1.227 \times 10^{-9} \text{ m} = 1.227 \times 10^{-7} \text{ cm}$

Now compare with the options :

- A. 2.48×10^{-8} cm $\approx 2.48 \times 10^{-10} \text{ m}$ $\nrightarrow$ too small
- B. 5.92×10^7 cm/sec $\nrightarrow$ units are velocity , wrong
- C. 8.66×10^{-6} cm $\nrightarrow$ too large
- D. 3.14×10^{-5} cm $\nrightarrow$ too large
- E. 9.81×10^{-4} cm $\nrightarrow$ too large
- F. $5.27 \times 10^{-9} \text{ m} \approx 5.27 \times 10^{-11} \text{ cm}$ $\nrightarrow$ incorrect (should be $\sim 1.23 \times 10^{-7} \text{ cm}$)
- G. $1.23 \times 10^{-7} \text{ cm}$ $\nrightarrow$ matches $1.23 \times 10^{-7} \text{ cm}$ $\nrightarrow$ correct
- H. $4.14 \times 10^{-10} \text{ m} \approx 4.14 \times 10^{-12} \text{ cm}$ $\nrightarrow$ wrong
- I. $6.625 \times 10^{-27} \text{ erg} \cdot \text{sec}$ $\nrightarrow$ this is h in erg · sec , not wavelength
- J. $0.512 \times 10^6 \text{ eV}$ $\nrightarrow$ energy unit , not wavelength

So , the correct answer is G : $1.23 \times 10^{-7} \text{ cm}$

Note : The value $1.23 \times 10^{-7} \text{ cm}$ is equivalent to $1.23 \times 10^{-9} \text{ m}$, which is the correct de Broglie wavelength of a 1 eV electron .

< answer > G < /answer >